

4.3 MONEY SUPPLY DEFINITIONS: M1, M2 & M3

Multiple Choice Questions

1990/CE/II/32

If Mr. Wong transfers HK\$100 from his savings account to his current account,

- A. the total amount of bank deposits will increase.
- B. the total money supply as measured by M1 will decrease.
- C. the total money supply as measured by M2 will not change.
- D. the total money supply as measured by M3 will increase.

1991/CE/II/34

Which of the following is a component of Hong Kong's money supply M2?

- A. savings deposits with licensed banks
- B. time deposits with deposit-taking companies
- C. negotiable certificates of deposit held by banks
- D. one-thousand-dollar gold coins issued by the Hong Kong government and held by banks

1992/CE/II/32

In Hong Kong, savings deposits with licensed banks are

- A. legal tender.
- B. included in the money supply M1.
- C. included in the money supply M2.
- D. included in the money supply M2, but excluded from the money supply M3.

1994/CE/II/43

If an individual receives a cheque for \$1 000 from his friend and deposits it in his savings account, Hong Kong's money supply M1 will _____ and M2 will _____.

- A. increase; increase
- B. remain unchanged; increase
- C. decrease; remain unchanged
- D. remain unchanged; remain unchanged

1995/CE/II/43

Mr. Chan withdraws \$600 000 from his savings deposit with a commercial bank and then deposits \$500 000 with a restricted licence bank. The remaining \$100 000 is kept at home. The immediate effect is that

- A. M1 will increase by \$100 000.
- B. M2 will decrease by \$600 000.
- C. M2 will not change.
- D. M3 will increase by \$500 000.

1996/CE/II/37

Mr. Chan paid \$200 000 in cash to buy from Mr. Wong a certificate of deposit issued by a deposit-taking company. Mr. Wong then deposited this \$200 000 into his savings account. The immediate effects are

- (1) M1 decreases by \$200 000
 - (2) M2 remains unchanged
 - (3) M3 increases by \$200 000
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

1997/CE/II/35

In Hong Kong, a demand or savings account holder can use an EPS card to pay for his purchases in certain shops. The function of an EPS card is that the amount of money is immediately transferred from the customer's bank account to the shop's bank account.

The popular use of EPS in Hong Kong will

- A. increase the ability of credit creation of the banking system.
- B. increase the buyer's cost of purchasing goods and services.
- C. increase the money supply M1 in Hong Kong.
- D. enable all bank deposits to act as a medium of exchange.

1997/CE/II/36

Which of the following will increase M1 in Hong Kong?

- A. a repayment of HK\$5000 by cheque on a bank loan in Hong Kong
- B. the lending of HK\$5000 in cash by a bank to a citizen in Hong Kong
- C. a new deposit of HK\$5000 in a savings account with a bank in Hong Kong
- D. a withdrawal of HK\$5000 in cash from a current account with a bank in Hong Kong

1997/CE/IV/37

Suppose the money supply data of Hong Kong is as follows:

	\$ billion
Total issue of legal tender	3
Legal tender in circulation	2
Demand deposits	7
Deposits with licensed banks AND negotiable certificates of deposit issued by licensed banks (other than those held by authorised institutions)	10
Deposits with restricted licence banks and deposit-taking companies AND negotiable certificates of deposit issued by restricted licence banks and deposit-taking companies (other than those held by authorised institutions)	80

The money supply M2 is

- A. \$9 billion.
- B. \$12 billion.
- C. \$13 billion.
- D. \$92 billion.

1998/CE/II/35

In Hong Kong, the users of some services can use the Payment by phone Service (PPS) to pay their bills over the telephone. After registration, they can make a phone call and transfer money from their bank accounts (savings or current) to the bank account of the service provider.

When the use of PPS becomes more popular,

- A. The money supply M1 will decrease.
- B. The money supply M2 will decrease.
- C. The money supply M3 will decrease.
- D. The amount of savings or current deposit of banks will decrease.

1998/CE/II/36

Which of the following will cause an immediate change in the money supply of M1 of Hong Kong?

- A. You receive \$500 cash from your father and deposit it in your current account.
- B. You pay \$4 000 cash to the landlord, who puts the money in his safe.
- C. A fire burnt most of the banknotes kept in a bank.
- D. You buy \$10 000 worth of furniture with a cheque. The shop owner deposits it into his time deposit account.

1999/CE/II/39

Suppose in a bank in Hong Kong, Mary withdraws HK\$3 000 from her saving account and changes it for US dollars in cash. Which of the following statements about the Hong Kong dollar money supply is correct?

- A. HK\$ M1 will increase while HK\$ M2 will remain unchanged
- B. HK\$ M1 will remain unchanged while HK\$ M3 will decrease.
- C. Both HK\$ M1 and HK\$ M3 will remain unchanged.
- D. Both HK\$ M1 and HK\$ M2 will decrease.

2000/CE/II/27

Mr Wong withdraws \$500 000 in cash from a deposit-taking company. He then deposits \$300 000 with a bank as a time deposit and remits the remaining \$200 000 overseas for the settlement of debts. Which of the following about the money supply M1 and M3 is correct?

- A. Both M1 and M3 will increase.
- B. Both M1 and M3 will decrease.
- C. M1 will remain unchanged and M3 will decrease.
- D. Both M1 and M3 will remain unchanged.

2000/CE/II/28

Suppose the money supply of Hong Kong in a certain year was as follows:

	\$billion
Legal tender in circulation	10
Legal tender held by all deposit-taking institutions	2
Demand deposits with licensed banks	15
Savings and time deposits with licensed banks	60
Negotiable certificates of deposits issued by licensed banks and held by the public	25
Deposits with restricted licence banks and negotiable certificates of deposits issued by restricted licence banks held by the public	32

The money supply M2 (in \$ billion) was

- A. 85
- B. 110
- C. 112
- D. 144

2001/CE/II/25

In using the Easy Pay System (EPS) to settle a payment in exchange, an amount of money is immediately transferred from the buyer's bank account to the seller's. In response to an increase in EPS's service charge, some shops have stopped accepting payment by EPS.

Which of the following would be the immediate effect of the above change on the money supply M1 and M2?

- A. Both M1 and M2 will increase.
- B. M1 will increase and M2 will remain unchanged.
- C. M1 will increase and M2 will decrease.
- D. Both M1 and M2 will decrease.

2002/CE/II/33

'Currency in public circulation' and 'savings deposits' are both included in Hong Kong's money supply

- (1) M1.
 - (2) M2.
 - (3) M3.
- A. (2) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

2003/CE/II/32

Mr Chan withdraws HK\$50 000 from his savings account with a bank in Hong Kong and remits HK\$40 000 through a bank to the U.K. He changes the remaining HK\$10 000 into the newly issued HK\$10 notes and puts them into his safety deposit box in a bank.

Which of the following would be the immediate effect of the above actions on the Hong Kong dollar money supply?

- A. Both the HK\$ M2 and M3 will decrease by HK\$40 000.
- B. HK\$ M1 will increase by HK\$ 10 000 while HK\$ M2 will decrease by HK\$50 000.
- C. HK\$ M2 will decrease by HK\$50 000 while HK\$ M3 will remain unchanged.
- D. HK\$ M1 will remain unchanged while HK\$ M2 will decrease by HK\$40 000.

2004/CE/II/37

Which of the following would decrease the money supply M2 in Hong Kong?

- A. Mr Au repays a HK\$10 000 loan in cash to a licensed bank in Hong Kong.
- B. Mr Bean withdraws HK\$20 000 from a savings account with a licensed bank in Hong Kong and holds it as cash.
- C. Mr Chan transfers HK\$30 000 from his savings account to his current account with a licensed bank in Hong Kong.
- D. Mr Ding withdraws HK\$40 000 from his time deposit in a restricted licence bank in Hong Kong and remits the money overseas.

2005/CE/II/38

Suppose Mrs Wong withdraws HK\$300 000 from her savings deposit with a commercial bank. She converts HK\$200 000 of the above amount into Renminbi (人民幣) and deposits it into a Renminbi savings deposit with the same bank. She keeps the remaining HK\$ 100 000 at home. The immediate effect of the above events is:

- A. Hong Kong dollar money supply M1 would increase by \$100 000.
- B. Hong Kong dollar money supply M2 would decrease by \$300 000.
- C. Hong Kong dollar money supply M2 would remain unchanged.
- D. Hong Kong dollar money supply M3 would increase by \$100 000.

2006/CE/II/35

Mr Chan withdraws HK\$20 000 from his deposit in a restricted licence bank in Hong Kong. He later puts HK\$15 000 as time deposit in a licensed bank in Hong Kong and holds the remaining HK\$5 000 as cash.

The immediate effect of the above actions on the Hong Kong dollar money supply would be:

- A. Both M2 and M3 decrease.
- B. Both M1 and M2 remain unchanged.
- C. M1 increases while M2 remains unchanged.
- D. M1 increases while M3 remains unchanged.

2007/CE/II/39

If an individual deposits \$600 000 cash into a time deposit account with a deposit-taking company, Hong Kong's money supply M1 will _____ and M2 will _____.

- A. decrease decrease
- B. decrease remain unchanged
- C. remain unchanged increase
- D. increase increase

2008/CE/II/37

Suppose, after making a series of unsuccessful add-value transactions through EPS in which their bank accounts were debited, more Octopus cardholders use cash to make their add-value transactions. This would immediately increase

- A. the money supply M1.
- B. the money supply M2.
- C. the amount of savings deposits of banks.
- D. the amount of time deposits of banks.

2010/CE/II/39

Grace receives a remittance of HK\$530 000 from overseas. She saves HK\$500 000 of the total amount as time deposits in a restricted licence bank and the rest as demand deposits in a licensed bank. What will be the immediate effect of the above actions on the money supply of Hong Kong?

- A. M1 will increase by HK\$30 000 and M2 will increase by HK\$530 000.
- B. M1 will increase by HK\$30 000 and M3 will increase by HK\$500 000.
- C. M2 will increase by HK\$30 000 and M3 will increase by HK\$500 000.
- D. M2 Will increase by HK\$30 000 and M3 will increase by HK\$5 30 000.

2012/DSE/I/28

Winnie withdraws HK\$500 000 from her time deposit with a deposit-taking company. Then she deposits HK\$200 000 as a time deposit and HK\$100 000 as a demand deposit with a licensed bank. She keeps \$50 000 in a safe at home and remits the remainder overseas.

What will be the immediate effect of the above actions on the Hong Kong dollar money supply?

- A. M1 will increase by HK\$50 000 and M2 will increase by HK\$350 000.
- B. M1 will increase by HK\$150 000 and M3 will decrease by HK\$150 000.
- C. M2 will increase by HK\$150 000 and M3 will decrease by HK\$150 000.
- D. M2 will increase by HK\$350 000 and M3 will decrease by HK\$500 000.

2013/DSE/I/25

Which of the following is included in the money supply M2?

- (1) legal tender held by licensed banks
 - (2) demand deposits with licensed banks
 - (3) time deposits with restricted licence banks
 - (4) negotiable certificates of deposit issued by licensed banks held by the non-bank public
- A. (2) and (3) only
B. (2) and (4) only
C. (1), (2) and (4) only
D. (1), (3) and (4) only

2015/DSE/I/30

Which of the following transactions would increase the money supply (M2) of Hong Kong?

- A. Ms Chan transfers \$500 000 from her time deposit account in a restricted licence bank to her time deposit account in a licensed bank.
B. Mr Lee transfers \$200 000 from his current account to his savings account in a licensed bank.
C. Mr Man deposits \$100 000 cash into his savings account in a licensed bank.
D. Ms Chung deposits \$600 000 cash into her time deposit account in a deposit-taking company.

2016/DSE/I/32

Winnie withdraws HK\$500 000 from her savings account with a licensed bank. She puts half of it into her current account and deposits the other half at a deposit-taking company. Which of the following shows the immediate effects on the money supply of Hong Kong?

- A. M1 increases and M2 decreases.
B. M1 increases and M3 decreases.
C. Both M1 and M2 decrease.
D. Both M2 and M3 decrease.

2017/DSE/I/33

Refer to the following data about Hong Kong's money supply.

Components	\$ billion
Total issue of legal tender	300
Legal tender held by all authorised institutions	80
Demand deposits	100
All types of deposits with licensed banks and negotiable certificates of deposit issued by licensed banks (other than those held by authorised institutions)	3 000

The money supply M2 is _____.

- A. \$320 billion
B. \$3 180 billion
C. \$3 220 billion
D. \$3 300 billion

2019/DSE/I/31

When a Hong Kong citizen transfers \$500 000 from her time deposit account with a restricted licence bank to her savings account with a licensed bank,

- (1) money supply (M1) will increase,
- (2) money supply (M2) will increase.
- (3) money supply (M3) will increase.

- A. (1) only
- B. (2) only
- C. (3) only
- D. (2) and (3) only

2020/DSE/I/29

The legal tender in Hong Kong held by licensed banks is included in _____.

- (1) monetary base
- (2) money supply M2
- (3) money supply M3

- A. (1) only
- B. (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2022/DSE/I/41

Which of the following results **CANNOT** be a negative number?

- A. total exports *minus* total imports
- B. money supply (M2) *minuss* money supply (M1)
- C. factor income received from abroad *minus* factor income paid abroad
- D. GDP at market price *minus* GDP at factor cost

2023/DSE/I/33

Carmen withdraws HK\$600 000 from her savings deposits with a licensed bank in Hong Kong. She puts one-third of the amount into her current account and remits the remaining amount overseas.

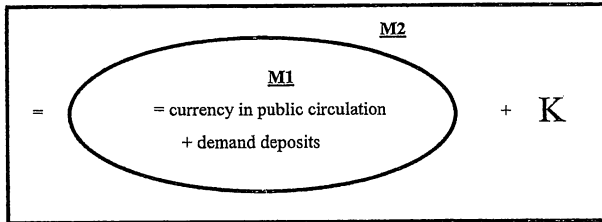
What will be the immediate effects of the above actions on the Hong Kong dollar money supply?

- (1) M1 increases by HK\$200 000.
- (2) M2 decreases by HK\$600 000.
- (3) M3 decreases by HK\$400 000.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2024/DSE/I/33

The following diagram shows the relationship between money supply M1 and M2 in Hong Kong.



In the above diagram, $M2 = M1 + K$. Which of the following are components of K?

- A. all types of deposits with licensed banks, restricted licence banks and deposit-taking companies
- B. time deposits with restricted licence banks and negotiable certificates of deposit issued by restricted licence banks held by non-bank public
- C. savings deposits and time deposits with licensed banks and negotiable certificates of deposits issued by licensed banks held by non-bank public
- D. reserves of licensed banks, restricted licence banks and deposit-taking companies

2025/DSE/I/32

Miss Ho uses HK\$500 000 cash to buy from Mr Lai a negotiable certificate of deposit issued by a deposit-taking company. Mr Lai then deposits HK\$200 000 cash to his Hong Kong dollar savings account and remits the remaining amount to his parents in Australia.

What will be the immediate effect of the above actions on the Hong Kong dollar money supply?

- A. M1 will increase while M3 will remain unchanged.
- B. M1 will remain unchanged while M3 will decrease.
- C. Both M1 and M3 will decrease.
- D. Both M1 and M3 will remain unchanged.

Short & Structured Questions

1991/CE/1/5(c)

- (i) Name **THREE** types of Hong Kong dollar deposits that a licensed bank in Hong Kong can accept. (3 marks)
- (ii) A person has a sum of money and is considering depositing it in a bank. Explain **TWO** factors that would affect his choice among the three types of Hong Kong dollar deposits. (6 marks)

1992/CE/1/3

- (a) (i) Which of the following components is/are included in the money supply M1 in Hong Kong? (2 marks)

Cash held by the non-bank public
Savings deposits
Demand deposits
Time deposits

- (ii) Explain why cash held by banks is not included in the calculation of M1. (3 marks)
- (b) In Hong Kong, Miss Wong puts \$1 000 cash into The Hongkong and Shanghai Banking Corporation Limited (HSBC) as a demand deposits.
- (i) What would be the immediate effect on M1? Explain. (3 marks)

1993/CE/1/5(c)(i)

Suppose Mr. Chan sells his flat to a friend in Canada and receives an overseas remittance which amounts to HK\$3 million. He deposits the HK\$3 million in his savings account with a bank in Hong Kong.

Explain the immediate effect on Hong Kong's money supply M₁ and M₂. (4 marks)

1995/CE/10(b)(ii)

In Hong Kong, a demand or savings deposit holder can use an EPS card to pay for his purchases in certain shops. The function of an EPS card is that the amount of money is immediately transferred from the customer's bank deposit account to the shop's bank deposit account.

Explain whether the money supply M₂ will be affected when a person uses an EPS card to buy goods. (2 marks)

1996/CE/1/7

The following are the consolidated balance sheets of all commercial banks for January and February in a certain year:

January:

Assets (\$mn)		Liabilities (\$mn)	
Cash Reserve	40	Demand Deposits	20
Loans	60	Savings Deposits	40
		Time Deposits	40

February:

Assets (\$mn)		Liabilities (\$mn)	
Cash Reserve	50	Demand Deposits	30
Loans	50	Savings Deposits	40
		Time Deposits	30

According to the above consolidated balance sheets, what are the changes in M₁ and M₂ from January to February? Show your calculation. (4 marks)

1999/CE/I/5

Define money supply M1 and M2.

(4 marks)

2002/CE/I/6

In Hong Kong, Mr Lee withdrew \$300 000 from his savings account in Bank A. He deposited 50% of this amount with a deposit-taking company and remitted through a bank another 20% of this amount overseas. He put the rest of the money in his safety box in Bank B.

What is the immediate effect of the above transactions on Hong Kong's money supply M1 and M3? Explain your answer.

(6 marks)

2008/CE/I/6

In Hong Kong, the money supply M2 consists of five components. One is given in the following figure. Fill in the other four components in the appropriate blanks.

(4 marks)

M2 =

M1 =

(1) _____

+

(1) _____

_____ placed with banks

+

Time deposits placed with banks

+

(3) _____

_____ placed with banks

+

(4) _____

_____ issued by banks held outside the banking sector

2020/DSE/I/8

Study the following balance sheet of the banking system in Country A .

Assets (\$million)		Liabilities (\$ million)	
Reserves	1 000	Deposits	4 000
Loans	3 000		

Suppose the public always hold \$800 million cash and the banking system does not hold excess reserves.

(a) Find the monetary base and money supply in Country A . (2 marks)

(b) Suppose firms withdraw \$700 million from the banking system and remit the whole amount overseas.

(i) Find the change in monetary base in Country A. (1 mark)

(ii) Calculate the maximum possible change in money supply in Country A. Show your workings. (3 marks)

2020/DSE/II/9C

Ocean Park and Hong Kong Disneyland are the two large theme parks in Hong Kong.

- (c) Instead of obtaining bank loans to finance the infrastructure development, it is suggested that Ocean Park can be listed in the stock market and issue shares to raise capital. State TWO advantages of issuing shares over obtaining loans from banks for raising capital. (2 marks)

2025/DSE/II/10(Cross Topic)

The monetary policy of the US receives much attention. (a) The table below shows some data on the US economy.

	May 2022	May 2023
Money supply (USD Billion)	21691.3	20 837.4
Inflation rate (measured by year-on-year percentage change in implicit price deflator of GDP)	7.0%	3.6%

- (i) Calculate the money supply growth rate in the US from May 2022 to May 2023. (1 mark)
- (ii) Based on the above data, what was the real output growth rate in the US from May 2022 to May 2023 as implied by the quantity theory of money? Explain your answer. (3 marks)

The Federal Reserve (the central bank of the US) began to lower the discount rate in September 2024.

- (b) **WITHOUT** using a diagram, explain how the equilibrium nominal interest rate would be affected by the lowering of the discount rate. (3 marks)
- (c) With the aid of an AS-AD diagram, explain how the real output in the US would be affected in the short run by the lowering of the discount rate. (5 marks)
- (d) To achieve the same effect on the real output as in part (c), an economist suggested that the US Government should increase its expenditure on infrastructure instead of having the Federal Reserve lower the discount rate. Discuss the suggestion by giving **ONE** argument **FOR** and **ONE** argument **AGAINST** it. (4 marks)